

ADAPT-VQE on Imipramine, CAS(6e,6o), 12 Qubits

A pool comparison that was not a pool comparison,
and a spin audit that inverted a verdict

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Abstract

Imipramine, a tricyclic antidepressant, is treated with ADAPT-VQE in the CAS(6e,6o)/6-31G active space on 12 qubits, following the Hamiltonian construction reported by Koziell-Pipe *et al.* in arXiv:2607.22468. The reference is CASCI, exact within the active space, and the Hamiltonian carries a validation receipt bound by SHA-256 to the code that produced it.

With the generalized `uccgsd` pool of 435 operators, ADAPT-VQE selects 58 operators, terminates on its own gradient threshold, and lands 0.0173 mHa above the exact active-space energy — recovering 99.92% of the 20.736 mHa of correlation available. Particle-number leakage is exactly zero, by construction rather than by penalty. The emitted Qiskit circuit, 2,988 two-qubit gates at depth 5,296, reproduces the optimised energy to 8.1×10^{-10} Ha with exact rather than Trotterised synthesis.

Two findings qualify that result, and both concern the validity of comparisons rather than the arithmetic.

First, the second run — the `uccsd` pool, 63 operators, presented as a comparison of operator pools — is not one. It halted on an exhausted operator budget at 31 selections with a residual pool gradient of 2.0×10^{-2} , twenty times its own stopping threshold. It was still descending. The two runs therefore differ in *when they were stopped*, not in what their pools could reach, and no conclusion about pool quality follows from their difference.

Second, that run’s energy of 0.892 mHa above CASCI would read as chemical accuracy. It is not usable: $\langle \hat{S}^2 \rangle = 2.8 \times 10^{-3}$ prices at 2.76 mHa against a 0.16 mHa budget, so the state is not the singlet the reference describes. One expectation value inverted the verdict.

Classical coupled cluster in the identical active space reaches 0.008 mHa at 502 seconds against ADAPT-VQE’s 0.017 mHa at 133 seconds. This is a verified reproduction, not a demonstration of advantage, and the classical ladder is reported so that the distinction is checkable rather than asserted.

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1 Introduction

1.1 Motivation

Imipramine is the closest thing at twelve qubits to a drug molecule with published quantum work at the same active space. It is a 45-atom tricyclic antidepressant, and the published study treats it three times — CAS(6e,6o), CAS(6e,7o) and CAS(8e,8o), at 12, 14 and 16 qubits. Those are separate calculations; this report concerns the first.

The purpose is a verified reproduction. The published statevector result is 1.15 mHa from CASCI using ADAPT-GQE circuits, and their 12-qubit training datasets were converged only to 5–10 mHa, so the target here is deliberately harder than the published one: 1.6 mHa, with symmetry accounting.

Table 1: The molecular system and the resulting qubit problem.

Molecule	Imipramine, 45 atoms
Geometry source	NIH PubChem CID 3696, conformer 00000E7000000001
Basis	6-31G
Total electrons / spatial orbitals	152 / 237
Frozen core orbitals	73
Active space	CAS(6e,6o), canonical RHF frontier orbitals
Mapping	Jordan–Wigner
Qubits	12
Pauli terms	1,819 (none discarded at 10^{-6} Ha)
FCI dimension in the active space	400
Nuclear repulsion	1660.4420186086 Ha
Core energy	−837.4744699644 Ha

Table 2: Reference energies for the cached Hamiltonian, by exact diagonalisation in the six-electron sector.

Quantity	Energy
$E(\text{RHF})$	−841.9973061558 Ha
$E(\text{CASCI})$ — the reference	−842.0180419414 Ha
Sector ground state (independent)	−842.0180419414 Ha
Correlation energy in the active space	20.736 mHa

1.2 System

1.3 Relationship to the QARP implementation

This workflow is a backend port. `molecule.py` and `validation.py` are byte-identical to those of the Fujitsu QARP / Qulacs implementation in `../imipramine_cas6e6o/`, and `hamiltonian.py` differs only in its docstring and two comments; `tests/test_core.py` fails if that ever stops being true. The consequence is that a disagreement between the two backends is a backend result and not a different molecule — which is the only condition under which comparing them would mean anything.

1.4 Scope and claims

1. ADAPT-VQE with the `uccgsd` pool attains 0.0173 mHa against exact diagonalisation of the identical cached Hamiltonian, with exactly zero particle-number leakage and $\langle \hat{S}^2 \rangle$ inside budget.
2. The emitted Qiskit circuit reproduces that energy, so the result is a circuit result and not only an algebraic one.
3. The `uccsd` run does not support a conclusion about pool quality, because it was stopped by an operator budget rather than by convergence.
4. That run’s energy is additionally invalidated by spin contamination, and the invalidation is detected by one expectation value.
5. Classical CCSD(T) in the identical active space outperforms both, which bounds what this benchmark can claim.

2 Methodology

2.1 Active-space construction and its guard

The active space is the six frontier orbitals of the canonical RHF solution, with 73 doubly occupied orbitals folded into a core energy. `prepare` measures the localisation of each selected orbital rather than assuming it, and refuses to build if the frontier window is delocalised away from the π system. The thresholds recorded in the result files are 0.8 for occupied orbitals and 0.65 for virtuals, under the keys `min_active_orbital_localization` and `min_active_virtual_localization`. Neither guard was overridden.

2.2 Validation protocol

`validate` diagonalises the cached operator exactly in the target sector and writes a receipt bound by SHA-256 to the cache, the molecular specification, the physics source and the penalty setting. Every run reported here was executed against the receipt digests listed in Appendix A. The receipt records `selects_target_sector: true` and `penalty_ground_electrons: 6.0`, i.e. the ground state of the operator actually being minimised lies in the six-electron sector.

ADAPT requires the *unconstrained* entry — `number=0,spin=0` — because its ansatz conserves particle number structurally and a penalty term would be inert overhead. That entry is what both runs consumed.

2.3 Acceptance criteria

A run is `verified_chemical_accuracy` only if all four hold:

1. $|E - E_{\text{CASC}}| < 1.6 \text{ mHa}$ for the identical cached Hamiltonian;
2. symmetry contamination, priced at 1 Ha per unit of $(\langle \hat{N} \rangle - 6)^2$ and per unit of $\langle \hat{S}^2 \rangle$, totals below 0.16 mHa;
3. the energy is not below the exact sector ground state by more than its own measured leak, which for a state inside the sector is impossible;
4. the emitted Qiskit circuit reproduces the optimised energy to 10^{-9} Ha.

Criterion (4) is specific to ADAPT and is not decorative. The optimisation runs in sparse linear algebra; the circuit is a separate object constructed afterwards. If `PauliEvolutionGate` synthesis were inexact — a Trotter step rather than an exact exponential — the energy would remain correct while the circuit would not reproduce it, and the run is reported as `CIRCUIT DISAGREES`. Every Pauli string within one excitation generator commutes with the others, so the synthesis is exact; this is measured rather than assumed.

2.4 Why ADAPT rather than a fixed ansatz

Three properties follow from the pool being fermionic excitation operators with balanced creation and annihilation indices.

Particle number is structural. Every pool operator commutes with $\hat{N}$ and $\hat{S}_z$ exactly, so the state cannot leave the six-electron sector. This is why `adapt` runs at zero penalty, why its leakage is exactly zero rather than merely small, and why the entire penalty and sector-selection apparatus that a hardware-efficient path requires does not apply.

S^2 is a weaker claim, and is not overstated. Spin-complementing each excitation with its partner makes the *singles* proper singlet operators, and the *pair* pool is seniority-zero, so both

Table 3: The two available methods in this workflow.

	run.py adapt	run.py vqe
Ansatz	grown by gradient	<code>real_amplitudes</code> , fixed
Particle number	exact by construction	broken, penalised, priced
S^2	measured and priced	measured and priced
Gradients	adjoint, $O(k)$	parameter shift, $O(k^2)$
Depth	an output	a knob
Engine	SciPy sparse algebra	Qiskit Aer circuits
Matches the paper	yes	no

commute with S^2 exactly. A general spin-complemented *double* does not: complementing buys invariance under flipping S_z , which is strictly weaker than commuting with S^2 . Measured directly from the commutators on the 12-qubit pools:

Pool	Singles breaking S^2	Doubles breaking S^2
<code>pair</code>	—	0/15
<code>uccsd</code>	0/9	27/54
<code>uccgsd</code>	0/15	315/420

Because the Hamiltonian is spin-free and the reference is a closed-shell singlet, the variational minimum *is* the singlet ground state, and $\langle \hat{S}^2 \rangle$ typically returns at 10^{-19} . But that is a property of the converged optimisation, not of the ansatz — which is precisely why $\langle \hat{S}^2 \rangle$ is measured at the optimum and priced, and why §3.3 exists.

Gradients are adjoint, and the optimisation is sparse algebra. Parameter shift needs $2k + 1$ evaluations for k parameters, each of which is itself k operator exponentials — $O(k^2)$. The reverse-mode sweep reuses every intermediate state and costs $2k$ exponentials in total. At twelve qubits the state is 4,096 amplitudes, so $\exp(\theta A)|\psi\rangle$ is a single `expm_multiply`, roughly a millisecond.

3 Results

3.1 The two runs

Table 4: Both ADAPT-VQE runs. Errors are against the CASCI reference of Table 2.

Pool	Ops	Error (mHa)	$\langle \hat{S}^2 \rangle$	2q gates	Depth	Status
<code>uccgsd</code> (435)	58	0.0173	1.10×10^{-6}	2,988	5,296	verified
<code>uccsd</code> (63)	31	0.8917	2.76×10^{-3}	2,515	3,424	spin contaminated

Table 5: Execution detail.

	uccgsd	uccsd
Pool size (singles / doubles)	435 (15/420)	63 (9/54)
Operators selected	58	31
Stop reason	pool gradient $8.62 \times 10^{-4} < 10^{-3}$	operator budget exhausted
Final pool gradient	8.62×10^{-4}	2.05×10^{-2}
Converged	yes	no
Matrix exponentials	29,817	9,188
Pool build time	37.4 s	14.0 s
Optimisation time	116.3 s	43.2 s
Total wall time	133.0 s	52.4 s
Circuit energy residual	8.11×10^{-10} Ha	6.78×10^{-10} Ha
Exact synthesis	yes	yes
Particle-number leakage	0.0 exactly	0.0 exactly

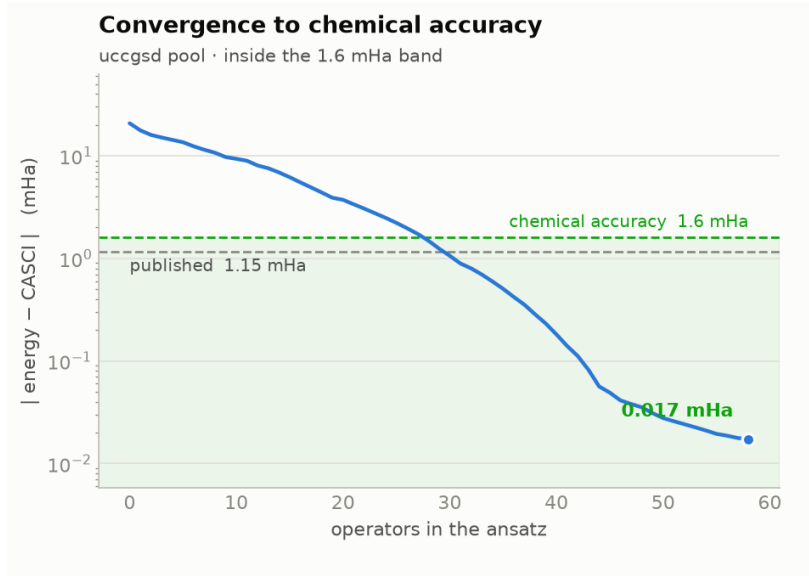


Figure 1: ADAPT convergence with the `uccgsd` pool: error against CASCI on a logarithmic axis, with the 1.6 mHa pass mark and the published 1.15 mHa statevector result drawn in. The trajectory crosses chemical accuracy well before it stops.

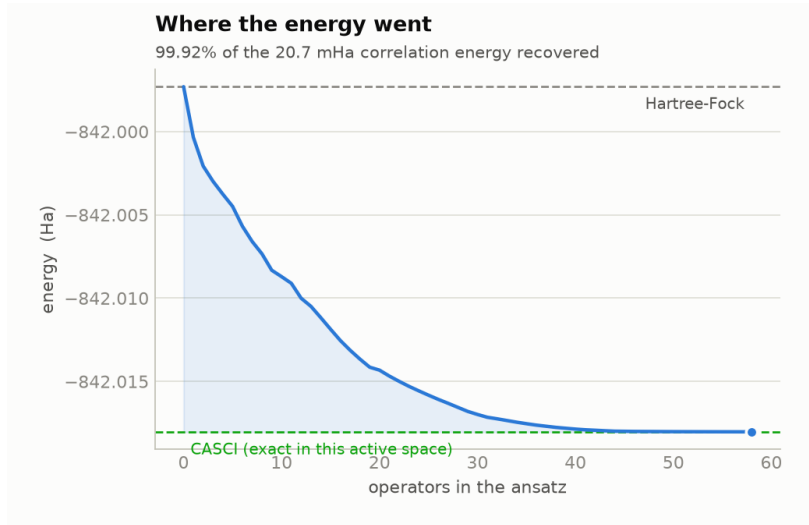


Figure 2: The same trajectory in absolute hartree, between the Hartree-Fock and CASCI lines, so the recovered correlation is a visible distance rather than a number to be trusted.

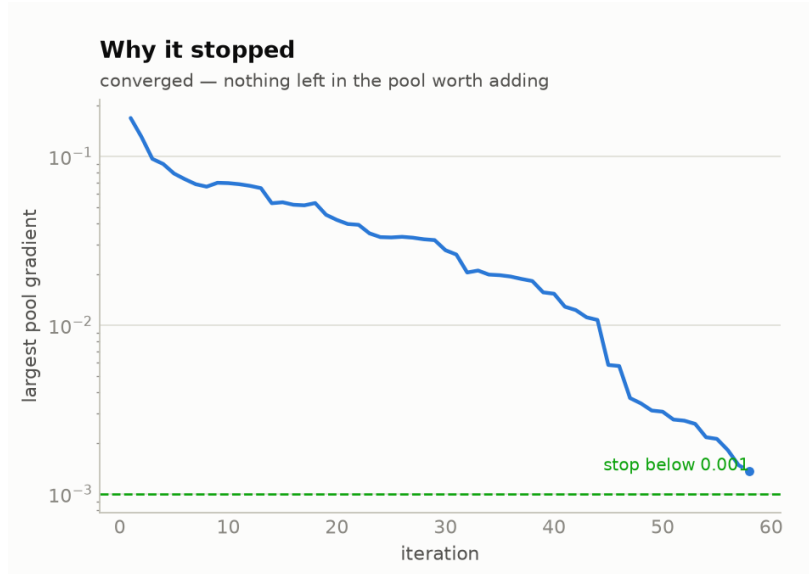


Figure 3: The largest remaining pool gradient by iteration — the termination criterion. This run reaches the threshold; Figure 5 does not, and the difference between those two figures is the subject of §3.2.

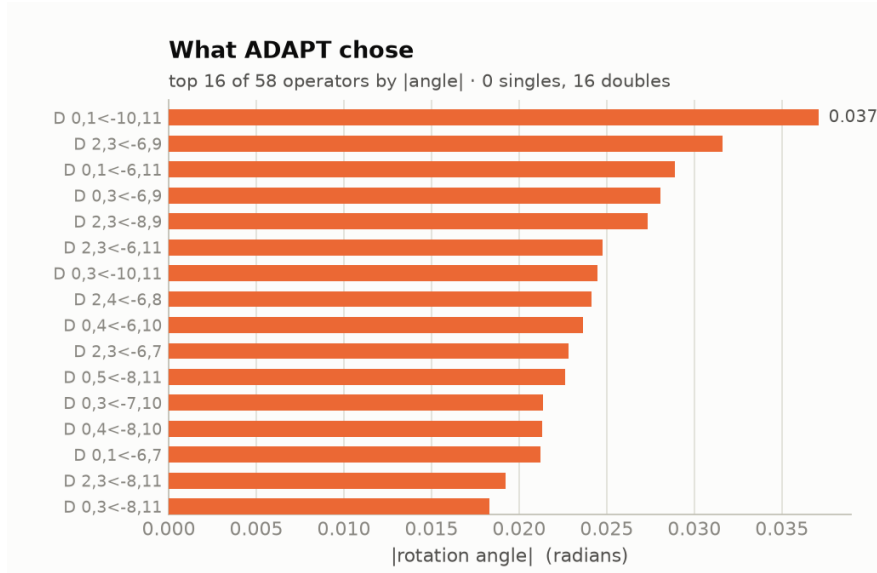


Figure 4: Selected operators by rotation angle, split into singles and doubles. The correlation is carried by a small number of large-amplitude doubles.

3.2 The uccsd run is not a pool comparison

The comparison that cannot be made

Table 4 invites the reading that the generalized pool is a factor of 52 more accurate than the singles-and-doubles pool. That reading is not supported. The uccsd run recorded `stop_reason: operator budget exhausted`, `converged: false`, and a final pool gradient of 2.05×10^{-2} against its threshold of 10^{-3} . It did not reach a stationary point. The two runs differ in **when they were stopped**, not in what their pools can express, and separating those two explanations requires re-running uccsd to its own gradient threshold with the operator cap raised.

This is a failure of experimental design, not of the algorithm, and it is visible only because `stop_reason` and `final_pool_gradient_max` are recorded in every result file. A result file carrying

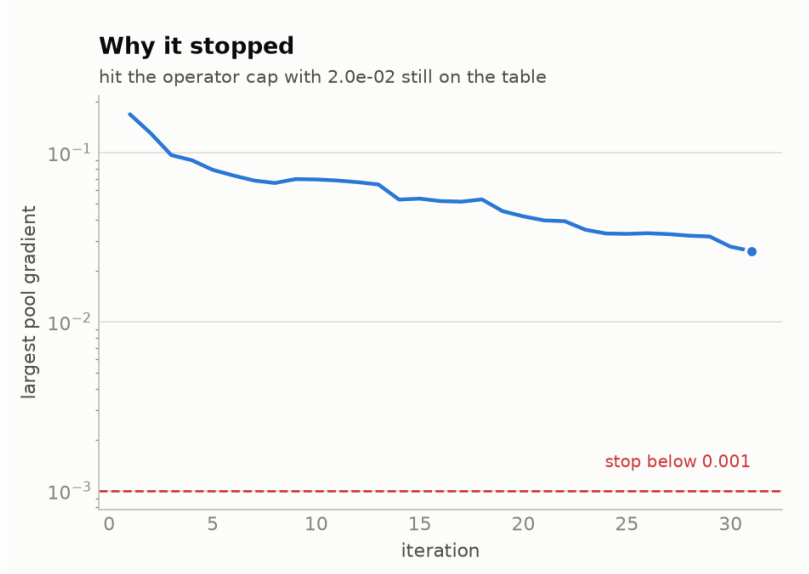


Figure 5: The uccsd run halting on its operator budget. The largest remaining pool gradient at termination is 2.05×10^{-2} — twenty times the stopping threshold of 10^{-3} . The curve has not flattened.

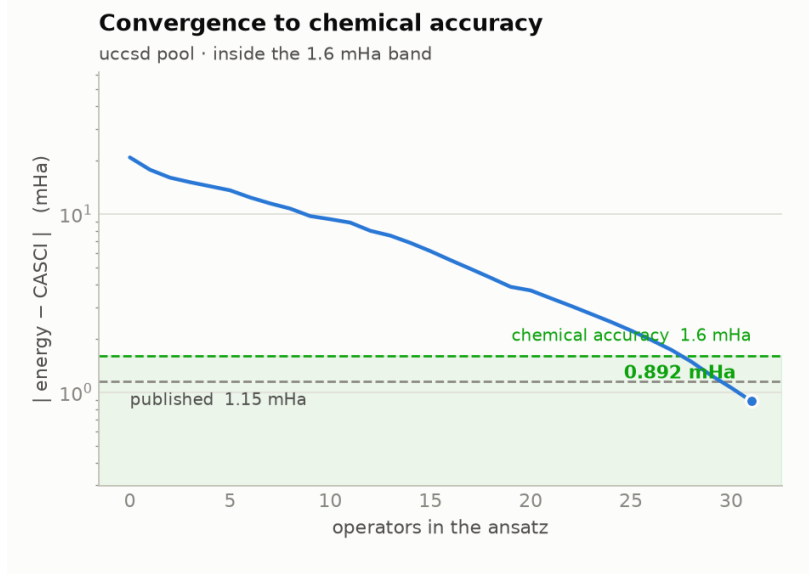


Figure 6: Error trajectory of the truncated run. It is still descending at the point where it was stopped.

only the final energy would have made the invalid comparison undetectable after the fact.

3.3 The spin audit

Table 6: Symmetry diagnostics at the converged state. *Priced* is the contamination energy at 1 Ha per unit, against the 0.16 mHa budget of §2.3.

Pool	$\langle \hat{N} \rangle - 6$	$\langle \hat{S}^2 \rangle$	Priced (mHa)	Verdict
uccgsd	0.0	1.10×10^{-6}	0.0011	inside budget
uccsd	0.0	2.76×10^{-3}	2.761	17× over

The uccsd run’s energy is 0.892 mHa above CASCI. Against a threshold of 1.6 mHa, that reads as chemical accuracy, and would be reported as such by any criterion that inspects the energy alone. It is not. The converged state carries $\langle \hat{S}^2 \rangle = 2.76 \times 10^{-3}$, which prices at 2.761 mHa — seventeen

times the contamination budget, and more than three times the error the run claims to have left. The state is not the closed-shell singlet that the CASCI reference describes, so the comparison between the two numbers is not between comparable objects.

What one expectation value bought

Measuring $\langle \hat{S}^2 \rangle$ on the converged state costs a single expectation value against a run of 9,188 matrix exponentials. It changed the verdict on that run from *chemical accuracy* to *not usable*.

Note the relationship between this finding and §3.2: they are the same failure seen twice. The run did not converge, and a non-converged ADAPT state is not obliged to be a spin eigenstate — S^2 symmetry is recovered at the variational minimum, and this run never reached one. The remedy for both is identical: raise `-max-operators`, or lower `-gradient-tolerance`, or use the seniority-zero pair pool whose doubles commute with S^2 exactly.

3.4 Circuit resources

	uccgsd	uccsd
Two-qubit gates	2,988	2,515
Single-qubit gates	4,674	1,960
Depth	5,296	3,424
Optimisation level	3	3
Synthesis	exact	exact

The circuit is constructed after the optimisation, transpiled at optimisation level 3 for a gate count, and its energy re-evaluated. Both runs agree with their algebraic energies below 10^{-9} Ha. These depths are far beyond current hardware coherence for a 12-qubit device; the numbers are reported as a resource estimate, not as a hardware proposal.

3.5 Classical baselines in the identical active space

Table 7: Classical methods frozen to CAS(6e,6o), so the quantum error has a scale. Errors against the CASCI reference.

Method	Energy (Ha)	Error (mHa)	Wall time (s)
RHF	-841.9973061558	20.736	92.5 (mean field)
MP2	-842.0102524898	7.789	5.7
CCSD	-842.0180213368	0.021	252.9
CCSD(T)	-842.0180339214	0.008	502.1
ADAPT uccgsd	-842.0180246890	0.017	133.0

What this benchmark can and cannot claim

CCSD(T) reaches 0.008 mHa of the exact active-space answer in 502 seconds; ADAPT-VQE reaches 0.017 mHa in 133 seconds. CAS(6e,6o) holds 400 determinants — small enough that coupled cluster is already excellent and exact diagonalisation is instant. **This is a verified reproduction, not a demonstration of quantum advantage.** The classical ladder is computed and reported so that the distinction is checkable rather than asserted.

The CCSD note in the result file is worth repeating: CCSD is not exact here, because a six-electron active space reaches hextuple excitations. The perturbative triples correction is -1.26×10^{-5} Ha, an order of magnitude smaller than the CCSD error itself, which is consistent with a weakly correlated active space.

3.6 Comparison with the published study

Published (arXiv:2607.22468, Table 1)	
CASCI energy	−841.953249 Ha
Statevector ADAPT-GQE	−841.952101 Ha
Statevector error	1.15 mHa
Hardware energy	−841.752841 Ha
Trained dataset accuracy	5–10 mHa

Compare differences, not totals

The published CASCI energy differs from the one here by 64.8 mHa. That is a conformer difference, not a disagreement: the geometry used here is a single PubChem conformer and the published work uses a private conformer dataset. The comparable quantity is $E_{\text{CASCI}} - E_{\text{HF}}$, i.e. the correlation energy captured in the active space, and the error against each study’s own reference. On that basis 0.017 mHa here against 1.15 mHa published is a like-for-like statement; the total energies are not.

4 Discussion

4.1 A single failure mode, seen twice

Both invalid comparisons in this study have the same shape: a quantity was compared against a reference that described a different object. In §3.2 the objects differed in operator budget; in §3.3 they differed in spin sector. In both cases the energies were computed correctly, the optimiser reported success, and the resulting numbers occupied a plausible range.

The instrumentation that caught them is three recorded quantities: the stop reason, the largest remaining pool gradient, and $\langle \hat{S}^2 \rangle$ at the optimum. None is expensive. The first two are free — the algorithm already computes them in order to decide when to halt — and the third is a single expectation value against a run of tens of thousands of matrix exponentials. What they cost is the discipline of writing them into the result file alongside the energy.

4.2 Why the accuracy test is energetic

Subtracting a penalty recovers $\langle \hat{H} \rangle$ exactly by linearity — but $\langle \hat{H} \rangle$ is a variational bound only *within* the six-electron sector, so a state that has leaked out of it can sit below CASCI while meaning nothing. Demanding zero leakage would fail every physically correct run of a hardware-efficient ansatz, which always leaks a little.

Pricing each broken symmetry at 1 Ha per unit and requiring the total below 0.16 mHa — one tenth of chemical accuracy — makes the criterion decidable and ensures that the contamination cannot be what produced the reported error. The price is a convention, stated so the budget can be recomputed under a different one.

4.3 The S^2 penalty is off by default, and that is not a weakening

Enabling the $\langle \hat{S}^2 \rangle$ penalty adds several hundred Pauli terms to every energy evaluation, to guard a secondary risk given a spin-free Hamiltonian, a real-amplitude ansatz and a closed-shell start. It is therefore off. But $\langle \hat{S}^2 \rangle$ is still *measured* at the optimum and still priced into the same budget at the full 1 Ha rate, so switching the penalty off cannot buy a weaker claim — as §3.3 demonstrates, since that verdict was produced with the penalty off.

5 Claims not supported by these data

1. **That uccgsd is a better pool than uccsd.** See §3.2. Deciding it requires re-running uccsd to its own gradient threshold.
2. **That 0.892 mHa is what the uccsd pool can reach.** It is where a truncated run stopped, and the state there is spin contaminated.
3. **Any claim about imipramine as a molecule.** A single PubChem conformer, in vacuum, in 6-31G, with six electrons in six orbitals. The active-space correlation energy is 20.7 mHa; the basis-set and active-space errors are not bounded by anything in this report and are certainly far larger.
4. **A comparison against the QARP/Qulacs backend.** The chemistry is pinned identical by test, which is the precondition for such a comparison, but the two backends were not run under matched settings here.
5. **Any hardware claim.** The circuits of §3.4 are resource estimates. Every energy in this report is a noiseless calculation, and the published hardware result for this system missed its own reference by 200 mHa.
6. **Quantum advantage, in any form.** Table 7 forecloses it at this size.

6 Limitations

1. One conformer, one geometry, one basis, one active space.
2. Two runs, one of which is invalid for two independent reasons. The pair pool was not run.
3. Seed 7; no statistics over initialisations. ADAPT is deterministic given the pool and the reference determinant — the result files record `deterministic: true` — so this is a smaller concern than for a randomly initialised VQE.
4. The results were produced on Colab with qiskit 1.0.2 and pyscf 2.14.0, which differs from the Windows environment the workflow also targets. Both are recorded in every result file.
5. The 1 Ha contamination price is a convention, not a derivation.

7 Conclusions

1. ADAPT-VQE with a generalized pool reaches 0.0173 mHa of exact diagonalisation for imipramine in the CAS(6e,6o) active space, on twelve qubits, in 133 seconds. It recovers 99.92% of the correlation energy available there, its particle-number leakage is exactly zero, and no penalty term was used.
2. The emitted circuit — 2,988 two-qubit gates, depth 5,296 — reproduces that energy to 8.1×10^{-10} Ha under exact synthesis, so the result is a circuit result.
3. The second run does not support a pool comparison. It halted on an operator budget with a residual gradient twenty times its threshold, and its energy is separately invalidated by a spin contamination that prices at seventeen times the budget. Both were identified by quantities the algorithm already computes.
4. Classical CCSD(T) in the identical active space is more accurate and slower; exact diagonalisation of 400 determinants is instant. This is a verified reproduction with classical controls, and it is reported as such.

A Provenance digests

Quantity	Digest (first 16 hex)
Molecular specification	b85747c4a6092698
Physics source	c6cd75127aac39a1
Chemistry workflow	c4682e8536f67d69
ADAPT workflow	afdc411eda4593dd
Hamiltonian cache	7a9d93ba991fe6c3
Receipt penalty key	number=0,spin=0

Both runs carry identical cache, specification and physics digests, so they solved the same operator.

B Complete settings

Setting	Value
Method	ADAPT-VQE
Backend	exact sparse statevector algebra (SciPy)
Gradient	adjoint (exact)
Inner optimiser	L-BFGS-B, tolerance 10^{-12} , max 500
Gradient tolerance	10^{-3}
Max operators	100 (uccgsd), 31 (uccsd)
Start	Hartree-Fock determinant, $\theta = 0$
Number penalty	0.0 Ha
Spin penalty	0.0 Ha
Contamination price	1.0 Ha per unit
Contamination budget	1.6×10^{-4} Ha
Hamiltonian cutoff	10^{-6} Ha
Circuit optimisation	level 3
Deterministic	true

The uccsd cap of 31 is the cause of §3.2.

C Software environment

Linux 6.6.122 (Google Colab), Python 3.12.13. qiskit 1.0.2, qiskit-aer 0.14.2, OpenFermion 1.8.1, PySCF 2.14.0, NumPy 1.26.4, SciPy 1.17.1.

The workflow also runs on Windows 11 / Python 3.12.13 with qiskit 2.5.2, qiskit-aer 0.17.2, OpenFermion 1.8.1, NumPy 2.5.2, SciPy 1.18.0 — every command except `prepare` and `classical`, which require PySCF and therefore a platform for which it publishes a wheel.

D Reproduction

On a machine with PySCF (Linux, macOS, WSL or Colab):

```
python run.py prepare
python run.py classical
```

Then anywhere:

```
python run.py validate --number-penalty 0 1 4
```

```
python run.py adapt --pool uccgsd
python run.py adapt --pool uccsd --max-operators 100    # the run that was
                                                         # truncated at 31

python run.py summary
python run.py plot
```

colab.ipynb performs prepare, validate and adapt in one free Colab session.

E Data inventory

File	Contents
hamiltonian_cas6e6o.json	the 12-qubit cache, 1,819 Pauli terms
hamiltonian_cas6e6o.validated.json	its receipt, three penalty keys
results/adapt_uccgsd.json	the verified run
results/adapt_uccsd.json	the truncated, contaminated run
results/classical.json	MP2, CCSD, CCSD(T) ladder
results/adapt_uccgsd.png	four-panel composite
results/adapt_uccsd.png	four-panel composite
results/comparison.png	cross-run leaderboard

Every quantity in this report is recomputable from those files. The composites in **results/** are the sources of the individual panels reproduced as Figures 1–6.

References

1. R. Koziell-Pipe *et al.*, arXiv:2607.22468 — the Hamiltonian construction reproduced here, and the source of the published comparison.
2. H. R. Grimsley *et al.*, *An adaptive variational algorithm for exact molecular simulations on a quantum computer*, Nat. Commun. **10**, 3007 (2019) — ADAPT-VQE.
3. National Center for Biotechnology Information, PubChem CID 3696, imipramine — the conformer used here.